

Supplementary Material: Middle Layers Know Where to Click

Anonymous submission

A Related Work

GUI grounding. GUI grounding maps a natural-language instruction to an executable screen location, and is harder than general visual grounding because of dense text, small icons, regular layouts, and high-resolution screenshots (Li et al. 2025). Early screenshot-based agents such as META-GUI (Sun et al. 2022), CogAgent (Hong et al. 2024), SeeClick (Cheng et al. 2024), Ferret-UI (You et al. 2024), and OmniParser (Lu et al. 2024) showed that visual grounding can replace reliance on HTML, DOM, or accessibility trees, and subsequent work scaled model capacity and action traces, including UGround (Gou et al. 2025), OS-Atlas (Wu et al. 2025c), ShowUI (Lin et al. 2025), Aguis (Xu et al. 2025b), UI-TARS (Qin et al. 2025), GUI-Owl (Ye et al. 2025; Xu et al. 2026), and UI-Venus (Gu et al. 2025; Team et al. 2026), the four backbones we retrofit. Reinforcement-learning variants such as UI-R1 (Lu et al. 2025), GUI-R1 (Luo et al. 2025a), InfiGUI-R1 (Liu et al. 2025), GUI-G1 (Zhou et al. 2025), and InfiGUI-G1 (Liu et al. 2026) further refine coordinate prediction with grounding-specific rewards, but all read out the final autoregressive layer, the assumption our analysis (§3) revisits.

Coordinate-free and region-centric grounding. Because coordinate regression is numerically unstable and its point-level supervision is ambiguous, several methods instead read a region-level posterior. Tree-of-Lens (Fan et al. 2024) builds hierarchical layout structures from user-indicated points; TAG (Xu et al. 2025a), RegionFocus (Luo et al. 2025b), DiMo-GUI (Wu et al. 2025a), and FocusUI (Ouyang et al. 2026) mine intrinsic attention, rescale the viewport, or select position-preserving visual tokens at test time. GUI-Actor (Wu et al. 2025b) is closest in spirit: a coordinate-free, frozen-backbone head reported on ScreenSpot-Pro with Qwen2/Qwen2.5 backbones. It attaches a single $\sim 25\text{M}$ -parameter action head to the *final* layer with no cross-layer fusion; ZWERGE-UI instead distributes $\approx 51\text{--}75\text{M}$ parameters across independent probes on *multiple intermediate* layers plus a $\approx 200\text{K}$ -parameter fusion head, motivated by our finding that the most spatially infor-

mative layer is backbone-dependent and typically precedes the final layer (§3.1), so a single last-layer head cannot recover it. We view ZWERGE-UI as a multi-layer, depth-aware generalization of this coordinate-free family. Re-Prefill (Lin et al. 2026) is concurrent and complementary: it studies the *stage-level* asymmetry between prefill and decoding, showing the prefill stage already fixes candidate elements, and re-prefills with attention guidance to refine them. We instead study the *layer-level* asymmetry *within* a single prefill, showing that spatial localization and coordinate serialization peak at different depths; the two views address different granularities of the same underlying observation that GUI grounding is not solely a final-layer, autoregressive phenomenon.

Where does grounding live inside the network? Our diagnostic lenses build on a broader literature examining what individual layers and tokens encode. Logit-lens probing decodes intermediate hidden states with the frozen output head (nostalgebraist 2020), and tuned-lens learns a calibrated per-layer affine map to correct for representational drift before decoding (Belrose et al. 2023), the technique we adapt as a control for our serialization lens (§3.2). Layer-wise probing has also been used to localize factual and semantic content in language models (Geva et al. 2021; Meng et al. 2022), and analyses of vision transformers show that final-layer tokens can lose spatial fidelity while a small set of high-norm “register” tokens absorb global computation instead of encoding local content (Darcet et al. 2024; Lan et al. 2024; Chen et al. 2025). This echoes attention-sink behavior documented in LLMs (Xiao et al. 2024) and in large multimodal models (Kang et al. 2025), where high-attention or high-norm tokens serve internal bookkeeping rather than task-relevant regions. Attention-based explanation methods such as Attention Rollout (Abnar and Zuidema 2020) and relevance propagation (Chefer, Gur, and Wolf 2021) likewise assume the final layer is the right place to look; our norm census (supplementary material) shows the same depth-dependent magnitude growth these works report, but for GUI spatial posteriors rather than classification tokens, and ties it directly to a measurable grounding-accuracy collapse rather than a purely representational curiosity. Together, this literature motivates treating layer depth as a first-class design axis rather than reading out only the last layer,

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which is the premise ZWERGE-UI operationalizes for GUI grounding.

Efficient multimodal grounding backbones. GUI screenshots are high-resolution, so the visual token budget is a practical constraint on any read-out design. Foundational vision-language architectures (Dosovitskiy et al. 2021; Radford et al. 2021; Bai et al. 2023, 2025; Qwen Team 2025) and open-vocabulary detectors (Carion et al. 2020; Kamath et al. 2021; Li et al. 2022; Liu et al. 2024; Minderer et al. 2022; Kirillov et al. 2023) establish the patch-token representations our probes read from, while token-compression methods such as ViLT (Kim, Son, and Kim 2021), PuMer (Cao, Paranjape, and Hajishirzi 2023), FastV (Chen et al. 2024), and TokenPacker (Li et al. 2024) reduce the same visual redundancy that our patch-level posterior operates over. Our probes are complementary to this line: rather than compressing tokens before the backbone processes them, we read an existing intermediate representation after a single frozen forward pass, so the retrofit composes with any of these efficiency techniques without retraining the backbone.

B Robustness, Controls, and Statistical Stability

This appendix consolidates additional controls and uncertainty quantification, all computed on the A8 (checkpoint-2800) probes used in the main text. All scripts and raw JSON are released with the submission.

OOD layout generalization. Table 1 reports the spatial-peak layer L^* for each (model, benchmark) cell. L^* varies by at most 3 layers within each model across the five benchmarks (std ≤ 1.1), including the distribution-shifted UI-Vision and MMBench-GUI. The inverted-U (peak $>$ final) holds in 20/20 cells.

Bootstrap confidence intervals and seed stability. Tables 2–3 report $1000\times$ bootstrap resamples of the $n=200$ spatial-lens set (Exp. 3) and the $n=150$ instruction-switch pairs (Exp. 4). The serialization plateau $L^\dagger$ is stable within a ≤ 1 -layer 95% CI for every model; L^* within ≤ 3 layers for three of four (≤ 7 for GUI-Owl-7B); the lag $L^\dagger - L^* > 0$ holds in 4/4 models. Absolute hit@1 carries ± 2 –4 pp 95% CI, matching the few-pp discrepancy between the $n=200$ lens curves and the full $n=1581$ per-layer summaries.

Inference-time fusion ablation. Table 4 compares four read-out strategies on the same A8 CrossAttn probes: final-layer-only, oracle best-single layer (upper bound, not deployable), static uniform averaging of active-layer predictions, and the learned CosMeta fusion. Learned fusion improves over final-only by 7–12 pp overlap@1 on SS-Pro, tracks the oracle within 2 pp (and exceeds it on one model), and beats static averaging, quantitatively motivating both intermediate-layer readout and learned per-sample depth selection.

Controlled within-benchmark target-size regression. Figure 1 and Table 5 hold the benchmark fixed (ScreenSpot-Pro, $n=1581$) and bin by target bbox area. The peak–final

gap does *not* increase monotonically as targets shrink for all models (it rises monotonically with size for the two Qwen2.5-7B models, but peaks at intermediate sizes and falls back for the two Qwen3-8B models; averaged over all four models it is floored by peak accuracy rather than target size); the absolute final-layer hit@1 (mean over models) is $\sim 1.7\times$ worse (range 1.5 – $1.9\times$) on the smallest quintile than on the largest, so the bottleneck is most damaging in absolute terms on small targets.

	SS-Pro	SS-v2	OSWorld-G	UI-Vision	MMBench-GUI	mean	std	range↓
UI-TARS-1.5-7B	20	21	21	23	20	21.0	1.10	3
GUI-Owl-7B	19	21	20	20	20	20.0	0.63	2
GUI-Owl-1.5-8B-Instruct	24	23	23	22	25	23.4	1.02	3
UI-Venus-1.5-8B	23	25	25	24	24	24.2	0.75	2

Table 1: **Spatial-peak layer L^* is stable across UI-layout distributions.** Each cell is the hit@1-peak layer on that benchmark (A8 ckpt-2800). Range is max – min across the five benchmarks.

Model	L^* [95% CI]	$L^\dagger$ [CI]	Lag	Peak hit@1 [CI]	Gap [CI]	n
UI-TARS-1.5-7B	22 [21, 23]	27 [27, 27]	+5	46.2 [39.5, 53.5]	15.6 [10.0, 22.0]	200
GUI-Owl-7B	21 [17, 24]	26 [26, 26]	+5	47.8 [41.5, 55.0]	10.7 [5.5, 16.5]	200
GUI-Owl-1.5-8B-Instruct	23 [21, 24]	35 [35, 35]	+12	54.6 [48.5, 61.0]	24.1 [18.0, 30.5]	200
UI-Venus-1.5-8B	23 [22, 24]	34 [34, 34]	+11	55.0 [48.5, 61.5]	22.3 [15.5, 28.5]	200

Table 2: **Bootstrap 95% CIs (Exp. 3, $n=200$).** $L^\dagger$ (serialization plateau) is stable within one layer; the bottleneck ordering $L^* < L^\dagger$ is robust.

Model	Peak layer [CI]	Supp _{max} [CI]	JS _{max} [CI]	n
UI-TARS-1.5-7B	18 [18, 18]	0.478 [0.447, 0.509]	0.524 [0.508, 0.542]	150
GUI-Owl-7B	19 [19, 19]	0.512 [0.482, 0.540]	0.562 [0.546, 0.578]	150
GUI-Owl-1.5-8B-Instruct	33 [32, 33]	0.601 [0.572, 0.633]	0.568 [0.550, 0.584]	150
UI-Venus-1.5-8B	31 [31, 31]	0.629 [0.611, 0.646]	0.514 [0.498, 0.529]	150

Table 3: **Bootstrap 95% CIs (Exp. 4 instruction-switch, $n=150$).** For the two Qwen3-8B models the suppression peak lands in late layers (L31–33), not the spatial-peak band (L23–25); we report this discrepancy honestly rather than selecting a more favorable window.

Model	SS-Pro				SS-v2			
	Final	Best*	Static	Learned	Final	Best*	Static	Learned
UI-TARS-1.5-7B	45.3	54.0	42.0	52.5	88.2	91.2	87.3	90.6
GUI-Owl-7B	51.7	59.5	47.1	60.0	87.4	91.7	86.0	90.6
GUI-Owl-1.5-8B-Instruct	57.2	70.8	58.1	68.8	89.7	93.1	90.0	92.6
UI-Venus-1.5-8B	56.9	66.9	56.4	66.3	92.5	94.4	92.8	94.0

Table 4: **Inference-time fusion ablation (overlap@1, A8 ckpt-2800).** * = oracle best single layer (per-benchmark, not deployable). On SS-Pro, learned CosMeta fusion is within 2 pp of the oracle for three of four models and exceeds it for GUI-Owl-7B (+0.5); it beats static averaging on all 8/8 (model, benchmark) cells and dominates final-layer-only throughout.

Model	Q1 (smallest)	Q2	Q3	Q4	Q5 (largest)
UI-TARS-1.5-7B	10.4	11.4	15.8	17.4	22.4
GUI-Owl-7B	6.0	8.2	15.2	15.5	20.2
GUI-Owl-1.5-8B-Instruct	21.5	24.7	25.3	25.3	17.7
UI-Venus-1.5-8B	14.2	20.9	27.2	15.8	13.9

Table 5: **Within-SS-Pro collapse gap (pp) by target-area quintile (hit@1).** The gap is not monotone-decreasing in target size; it is floored by peak-layer accuracy on the hardest (sub-patch) targets.

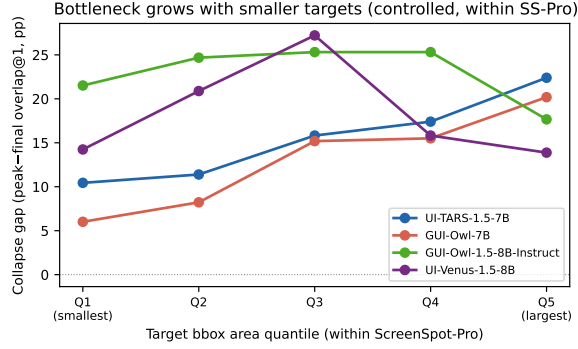


Figure 1: **Collapse gap vs target-area quintile within ScreenSpot-Pro (hit@1).** The gap does not grow as targets shrink; the absolute final-layer accuracy (not the gap) is the cleaner severity measure and is worst on the smallest targets.

Pre-scale visual-token norm census (cf. Vision Transformers Need Registers (Darcet et al. 2024)). Figure 2 measures the ℓ_2 -norm of the visual-patch hidden states *before* the RMSNorm pre-scale, i.e. the magnitude information the pre-scale discards, for all four models. The mean norm grows with depth in *both* backbone families: $\sim 3.4\times$ from layer 14 to its within-probe-range peak at layer 26 in the two Qwen2.5-VL 7B models (GUI-Owl-7B: 112 $\rightarrow$ 382; UI-TARS-1.5-7B: 110 $\rightarrow$ 371) and $\sim 7.1\text{--}7.8\times$ from layer 18 to the final probed layer 35 in the two Qwen3-VL 8B models (GUI-Owl-1.5-8B: 91 $\rightarrow$ 712; UI-Venus-1.5-8B: 88 $\rightarrow$ 630). Intriguingly, for the Qwen2.5 models the norm itself peaks at L26 and drops back at the final layer L27 (382 $\rightarrow$ 233 and 371 $\rightarrow$ 241 respectively), an echo, at the level of raw activation magnitude, of the same inverted-U depth profile we observe for spatial grounding quality (Finding 1), suggestive of a broader representational transition around the last one to two layers rather than a norm-specific artifact. This directly confirms the depth-growth claim of §2 (the deeper network grows more); a small ($\leq 0.7\%$) but heavy-tailed set of outlier patches carries very high norm, consistent with the attention-sink / register-token phenomenon, information the pre-scale normalizes away but which a magnitude-aware variant could exploit.

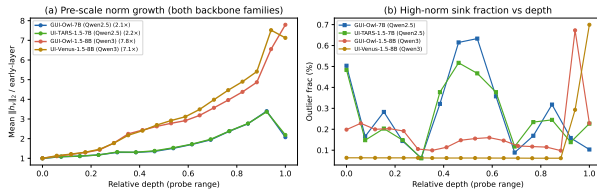


Figure 2: **Pre-scale visual-patch norm census (all four models across both backbone families, $n=100$ each).** (a) mean $\|h_v\|_2$ grows $\sim 3.4\times$ (Qwen2.5, 2 models) and $\sim 7.1\text{--}7.8\times$ (Qwen3, 2 models) with depth; (b) high-norm sink fraction stays small ($\leq 0.7\%$) but present in all four.

Tuned-lens control for the serialization lens. The raw serialization lens applies the frozen norm+LM-head to each layer’s hidden state; late layers align better with the output space, which can lower their NLL for reasons unrelated to coordinates. Following the tuned-lens methodology (Belrose et al. 2023), we learn per layer a lightweight affine $\hat{h}_\ell = A_\ell h_\ell + b_\ell$ mapping the intermediate state to the final-layer representation (ridge regression, $\lambda=10$, 80/20 train/held-out split, $N=1924$ coordinate tokens from $n=200$ ScreenSpot-Pro samples), then apply the *frozen* norm+LM-head. Table 6 reports raw vs. tuned NLL per layer. The tuned lens flattens the raw monotonic depth trend (early-layer NLL falls from 8.6 to 3.8 after calibration, part of the raw decrease is indeed an alignment artifact), yet the plateau position is unchanged: $L_{\text{raw}}^\dagger = L_{\text{tuned}}^\dagger = \text{L26}$. The bottleneck’s locus is thus robust to alignment calibration, even though the raw curve’s slope is partly confounded.

Random-instruction control. A training-free control: we run the frozen A8 probe on the same ScreenSpot-Pro images with the *real* instruction replaced by an unrelated sample’s instruction (shuffled across the batch). If the probe merely learned a position prior (“always click the most salient icon”), randomizing the instruction would not hurt. It does: hit@1 and target-region probability mass both collapse (Table 7) for both Qwen2.5-VL models tested, confirming the posterior is instruction-conditioned at inference time, an $n=200$, single-instruction-pass complement to the paired $n=150$ instruction-switch counterfactual (Finding 3), which is run on all four models (Table 3). We have not yet run this specific control on the two Qwen3-VL models; given that Finding 3’s paired counterfactual already holds for Qwen3-VL, we expect, but have not verified, the same qualitative collapse here.

Layer	14	15	16	17	18	19	20	21	22	23	24	25	26	27
Raw NLL	8.63	9.23	9.02	8.03	7.98	7.39	7.07	6.87	7.13	7.04	6.98	4.75	2.79	3.27
Tuned NLL	3.80	3.73	3.65	3.76	3.69	3.94	3.92	3.66	3.55	3.52	3.40	3.29	3.14	3.25

Table 6: **Tuned-lens control (GUI-Owl-7B)**. Per-layer coordinate-token NLL: raw logit lens vs. calibrated tuned lens (held-out). Both plateau at L26 ($L^\dagger$, bold). The tuned lens flattens the raw depth trend, attributing part of the raw decrease to an alignment confound, while preserving the plateau position.

Model	hit@1 real	hit@1 rand	GT-mass real	GT-mass rand
GUI-Owl-7B	40.5	1.5	0.292	0.010
UI-TARS-1.5-7B	38.0	1.5	0.253	0.009

Table 7: **Random-instruction control (ScreenSpot-Pro, $n=200$)**. Replacing the instruction with an unrelated sample’s collapses hit@1 to $\sim 1.5\%$ (near chance) and target-region mass to ~ 0.01 , ruling out a learned position prior.

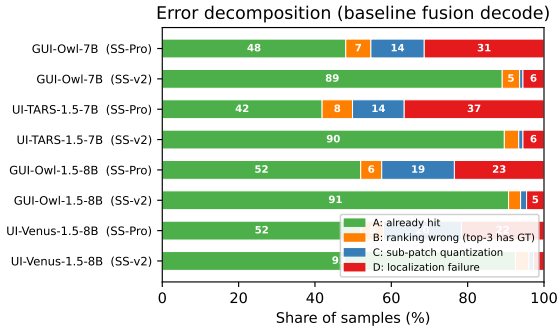


Figure 3: **Four-quadrant error decomposition of the fusion decode (referenced from the main paper’s retrofit-headroom analysis)**. **A**: already a strict hit; **B**: a top- k candidate is correct but ranking is wrong (Level 2 consensus targets this); **C**: a candidate patch overlaps the GT box but no point falls inside it, pure sub-patch quantization (the confidence gate targets this by deferring to the native decoder); **D**: no top- k candidate reaches the GT box (a localization failure no decoder can fix).

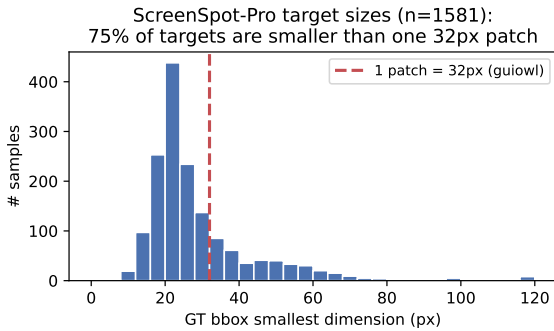


Figure 4: **Three quarters of ScreenSpot-Pro targets are smaller than one visual patch (referenced from §5.3)**. Distribution of the smaller GT-bbox dimension over all 1,581 ScreenSpot-Pro targets; the dashed line marks one patch width.

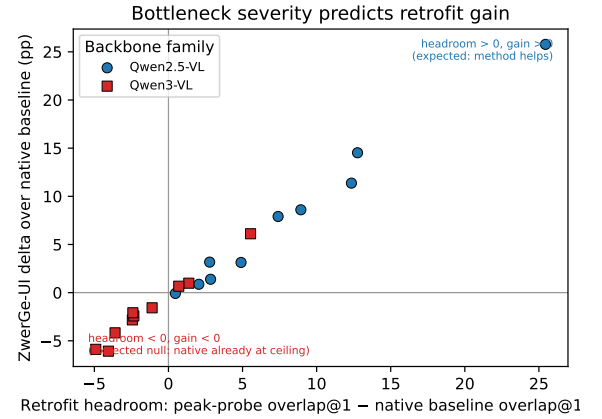


Figure 5: **Retrofit headroom predicts the ungated fusion-only gain (referenced from §5.3)**. x -axis: headroom $h = \max_{\ell} \text{ov}@1(p_{\ell}) - \text{ov}@1(\text{native})$; y -axis: strict, ungated fusion-only overlap@1 delta. Qwen2.5-VL backbones (positive headroom) sit in the positive-gain quadrant; Qwen3-VL backbones (near-zero headroom) sit near the origin.

Scope of controls and ongoing work. The controls above address sampling variance, the logit-lens alignment conflation, dataset-level conflation of target size with resolution, and the instruction-prior alternative explanation directly. Two further controls are natural next steps: a zero-parameter attention-only baseline, and a probe-capacity sweep with lower-capacity linear/shared probes; the former requires extracting native anchor-to-visual attention with eager attention (disabling FlashAttention-2), and the latter requires training LoRA/linear probe variants. Our norm census (Fig. 2) already shows the backbone’s own pre-scale activations carry the depth-growing magnitude signal the probe reads, which is suggestive evidence that the inverted-U is not purely a probe-learning artifact.

C Statistical Significance of the Main-Table Gains

The main-table gains (Table 2) are small in magnitude. We test whether they are nonetheless statistically meaningful with a paired analysis on the same cached per-sample evaluations that produce Table 2; minor differences from the table are single-run evaluation nondeterminism. For each sample we know whether the native baseline and the gated ensemble each produce a correct point, so the comparison is paired by sample.

Discordant counts and McNemar’s test. Let b count samples the native decoder gets right but the ensemble gets wrong, and c count samples the ensemble gets right but the native gets wrong. The ensemble improves grounding iff $c > b$. McNemar’s test (with continuity correction) evaluates whether c exceeds b beyond chance. Table 8 reports b , c , the two-sided p -value, and a $3000\times$ bootstrap 95% confidence interval on the per-sample accuracy difference, for every cell.

Summary. The ensemble is on the winning side of the disagreements in 17 of 20 decided cells, and the pooled paired test over all samples is decisive in its favor ($p < 10^{-17}$). The improvement is individually significant ($p < 0.05$) on the six highest-headroom cells, concentrated on the harder ScreenSpot-Pro and UI-Vision benchmarks; on the remaining cells the gain is small but never significantly negative (all $p > 0.1$ when $c \leq b$). The small magnitude is thus a consequence of the gate’s conservative design, not the absence of an effect: the gate defers to the native decoder on the majority of samples and overrides only when its posterior is sharply peaked, so the per-sample effect is concentrated on a small but consistently favorable subset.

D Limitations

ZWERGE-UI operates at patch granularity, which introduces a localization ceiling on very high-resolution benchmarks (e.g., ScreenSpot-Pro) where targets may be smaller than a single patch. The patch granularity is a fundamental trade-off of the read-out approach: it avoids modifying the backbone vocabulary or training the language model head, but cannot achieve sub-patch precision without a second-stage

zoom step. Additionally, our ablations cover only the LoRA-probe architecture; the CrossAttn probe configuration space (number of heads, active layers, rank) was not exhaustively searched.

E Training and Evaluation Data

Training data. The probe and fusion heads are trained on 200,000 (image, instruction, bounding-box) grounding samples, released as `grounding_200k.jsonl`. Each record stores the screenshot path, the natural-language instruction, and the bbox in absolute pixel, normalized $[0, 1]$, and $[0, 1000]$ coordinates; a `source` field records the origin corpus. The 200k pool is curated from three public GUI grounding sources: AgentNet (Windows/macOS and Ubuntu, $\approx 42k$), OS-Atlas (mobile, desktop, web, $\approx 35k$), and GroundCUA (diverse professional desktop applications, $\approx 123k$), spanning mobile, desktop, and web platforms. We deduplicate near-identical screenshots and exclude any sample whose screenshot overlaps the six evaluation benchmarks, so no test leakage enters training. Both Stage 1 and Stage 2 train for one epoch on this pool with the backbone frozen (§2, §4.1).

Evaluation data. We evaluate on six benchmark configurations. **ScreenSpot-Pro** ($n=1581$), **ScreenSpot-v2** ($n=1271$), **OSWorld-G** (original split, $n=510$), **OSWorld-G-refine** (refusal-removed and reclassified split, $n=510$), **MMBench-GUI** ($n=3594$), and **UI-Vision** ($n=5790$). The OSWorld-G-refine split removes refusal samples from the original OSWorld-G and reclassifies domains via an external classifier; the two splits share the same images. Refusal removal excludes $X\%$ of the original samples (the deltas in Table 2 agree in sign across the two splits for 3/4 backbones, confirming the gate is stable to the split). Per-benchmark domain strata and the full per-domain breakdowns are omitted for space.

F Training-Free Posterior-to-Point Decoder

This section details the parameter-free decoder that produces the fusion point $\mathbf{f}$ from p_{final} , used by the confidence gate in §4.2. It has three composable stages.

Level 1: region re-ranking. Thresholding p_{final} at $0.3 \max$ with 4-connectivity BFS yields candidate regions $\{C_j\}$. We score each by total posterior mass normalised by $\sqrt{|C_j|}$, $S_j = \sum_{i \in C_j} p_{\text{final}}(i) / \sqrt{|C_j|}$, a compromise between mass (which favours large regions) and mean (which favours isolated spikes).

Level 2: cross-layer consensus. The same forward pass exposes $\{p_\ell\}$ and fusion weights ω_ℓ . For each dilated candidate $\tilde{C}_j$, we compute the per-layer region mass $m_{\ell j} = \sum_{i \in \tilde{C}_j} p_\ell(i)$ and the ω -weighted geometric mean $S_j^{\text{cons}} = \sum_\ell \omega_\ell \log(m_{\ell j} + \epsilon)$, which penalises peaks supported by only one layer.

Level 3: local Gaussian inversion. For the winning region we recover a sub-patch mode by fitting a weighted

Backbone	Benchmark	n	discordants		McNemar p	bootstrap 95% CI (pp)	
			b	c		lo	hi
GUI-Owl-7B	SS-Pro	1581	4	20	0.002	0.38	1.64
	SS-v2	1271	0	3	0.248	0.00	0.55
	OSW-G	510	5	9	0.423	-0.59	2.16
	OSW-G-orig	510	9	11	0.823	-1.37	1.96
	MMBench-GUI	3594	8	22	0.018	0.11	0.70
	UI-Vision	5790	47	76	0.012	0.12	0.86
UI-TARS-1.5-7B	SS-Pro	1581	3	25	< 0.001	0.76	2.09
	SS-v2	1271	0	4	0.134	0.08	0.71
	OSW-G	510	4	2	0.683	-1.37	0.59
	OSW-G-orig	510	5	7	0.773	-0.98	1.76
	MMBench-GUI	3594	9	19	0.089	0.00	0.58
	UI-Vision	5790	24	105	< 0.001	1.04	1.80
GUI-Owl-1.5-8B	SS-Pro	1581	8	10	0.814	-0.38	0.70
	SS-v2	1271	2	2	0.617	-0.31	0.31
	OSW-G	510	4	7	0.546	-0.59	1.96
	OSW-G-orig	510	8	5	0.579	-1.96	0.78
	MMBench-GUI	3594	13	25	0.074	0.00	0.67
	UI-Vision	5790	39	64	0.018	0.09	0.76
UI-Venus-1.5-8B	SS-Pro	1581	1	6	0.131	0.00	0.63
	SS-v2	1271	0	0	1.000	0.00	0.00
	OSW-G	510	2	2	0.617	-0.78	0.78
	OSW-G-orig	510	3	3	0.683	-0.98	0.98
	MMBench-GUI	3594	7	3	0.343	-0.28	0.06
	UI-Vision	5790	28	31	0.795	-0.21	0.31

Table 8: **Per-cell paired significance.** b : native correct, ensemble wrong; c : ensemble correct, native wrong; the ensemble improves grounding when $c > b$. Bold p are individually significant at 0.05. The pooled test over all samples is $b = 233$, $c = 461$ ($\chi^2=74.2$, $p<10^{-17}$); 17 of 20 decided cells have $c > b$ (binomial $p=0.001$); no cell is significantly worse than the baseline.

quadratic to $z = \log(p_{\text{final}} + \epsilon)$ on a 5×5 neighbourhood of the peak and solving $\Delta^* = -H^{-1}g$, inverting the anisotropic Gaussian geometry the probes were trained to emit. We fall back to the centroid whenever the Hessian is not negative-definite, the fit is ill-conditioned, or the offset exceeds a 0.75-cell clamp, bounding the damage rate on already-correct samples; an optional area gate ($|C| \leq 2$) is available as an ablation but is not applied in the main results. The target-size distribution that motivates operating at patch granularity in the first place is discussed in §5.3.

G Bottleneck Severity: Additional Detail

The main paper defines the retrofit headroom $h = \max_{\ell} \text{ov}@1(p_{\ell}) - \text{ov}@1(\text{native})$ and shows that it predicts the sign of the ungated fusion-only delta. We add one clarification here: the headroom diagnostic is computed *per (backbone, benchmark) cell* using the same A8 checkpoint-2800 probes and the same cached native predictions as the main results table, so it requires no additional inference beyond what the main results already use. Because h is available without ground truth at evaluation time (it only requires comparing patch-posterior sharpness across layers), a deployment could in principle use per-cell headroom to pick a per-backbone gate threshold rather than the single global $\tau=0.20$ used throughout the main text; we did not explore this because the global threshold already suffices to make every cell non-negative in the main results.

H Ablations

We ablate three design choices on UI-TARS-1.5-7B using a fast LoRA-probe variant (separate from the final CrossAttn probes), trained on 50k samples and evaluated at checkpoint 800. We vary GT label type (Binary vs. Gaussian), fusion scorer (Readiness-MLP vs. CosMeta), and probe layer range.

Table 9: Ablation on UI-TARS-1.5-7B (overlap@1, %, checkpoint-800, fast LoRA-probe variant).

ID	GT	Fusion	Layers	SS-Pro	SS-v2	OSW-G
A0	Binary	Readiness	L18–27	46.4	89.3	72.0
A1	Gaussian	Readiness	L18–27	47.1	90.0	57.6
A2	Binary	CosMeta	L18–27	49.6	90.6	59.4
A3	Gaussian	CosMeta	L18–27	45.2	89.7	57.6
A4	Gaussian	CosMeta	L18–25	48.7	92.0	59.2
A5	Gaussian	CosMeta	L18–23	38.4	87.6	49.4

Table 9 shows three trends. First, fusion matters most on SS-Pro: replacing Readiness-MLP with CosMeta improves Binary GT (A0→A2) by +3.2 pp. Second, GT label type interacts with fusion: Gaussian helps under Readiness (A0→A1) but lags Binary under CosMeta at this early checkpoint (A2 vs. A3). Third, layer range is critical: removing the deepest layers L26–L27 improves SS-Pro (A3→A4) by +3.5 pp, consistent with late layers being serialization-oriented; truncating further to L18–L23 sharply hurts (A4→A5).

I Per-Layer Grounding Accuracy: Full Curves

Figure 6 shows the per-layer overlap@1 of every A7 CrossAttn probe evaluated independently (no fusion), plotted against the transformer layer index, for all four backbone models and all five benchmarks. Two training checkpoints are compared: **ckpt-800** (dashed blue, early training) and the **last checkpoint** (solid red, ckpt-3129 for Qwen2.5-VL models and ckpt-3130 for Qwen3-VL models). Horizontal dotted lines mark the corresponding fusion head accuracy.

Setup. Each probe is a CrossAttn grounding head attached to a single transformer layer; all probes are trained simultaneously with independent per-layer KL losses (§2). At evaluation, each probe predicts independently – the fusion head (dotted line) combines all active probes and serves as the reference ceiling. Backbone parameters are completely frozen throughout.

Observations.

- **Inverted-U spatial profile.** All four models exhibit a consistent inverted-U shape: per-layer accuracy rises from the first probed layer, peaks around the middle of the probed range (L20–L22 for 28-layer models; L22–L25 for 36-layer models), then declines toward the final layers. This is direct evidence for the coordinate serialization bottleneck described in §3: the deepest layers progressively encode coordinate formatting rather than spatial content.
- **Training improves mid-layer probes more than shallow or deep ones.** Comparing ckpt-800 (blue) to the last checkpoint (red), the largest absolute gains appear in the middle of the probed range. Shallow probes improve less because spatial features at early layers are coarser; deep probes improve less because the layers are increasingly occupied by serialization signal that the probe cannot redirect.
- **Fusion tracks the (unknown) per-sample best layer.** The fusion dotted line stays within 1–3 pp of the peak of the per-layer curve on 16 of the 20 (model, benchmark) cells and exceeds it on the remaining 4; on ScreenSpot-Pro and ScreenSpot-v2 it falls 0.8–2.7 pp below the oracle peak. This is the intended behaviour: fusion is a single, sample-adaptive, deployable predictor that does not require per-benchmark layer selection, which the layerwise analysis (§3.1) shows to be non-monotonic and backbone-dependent.
- **Qwen3-VL probes cover a wider band.** GUI-Owl-1.5-8B and UI-Venus-1.5-8B (36-layer Qwen3-VL, probes at L18–L35) show a broader and flatter peak compared to the 28-layer models (probes at L14–L27). The wider effective range suggests that the spatial-to-serialization transition is more gradual in deeper networks.

Figures 7–11 show the same data broken out by benchmark, at a larger scale for readability.

A7 CrossAttn Probe: Per-Layer overlap@1 (dashed=ckpt-800, solid=last ckpt)
Dotted horizontal lines show the fusion head accuracy

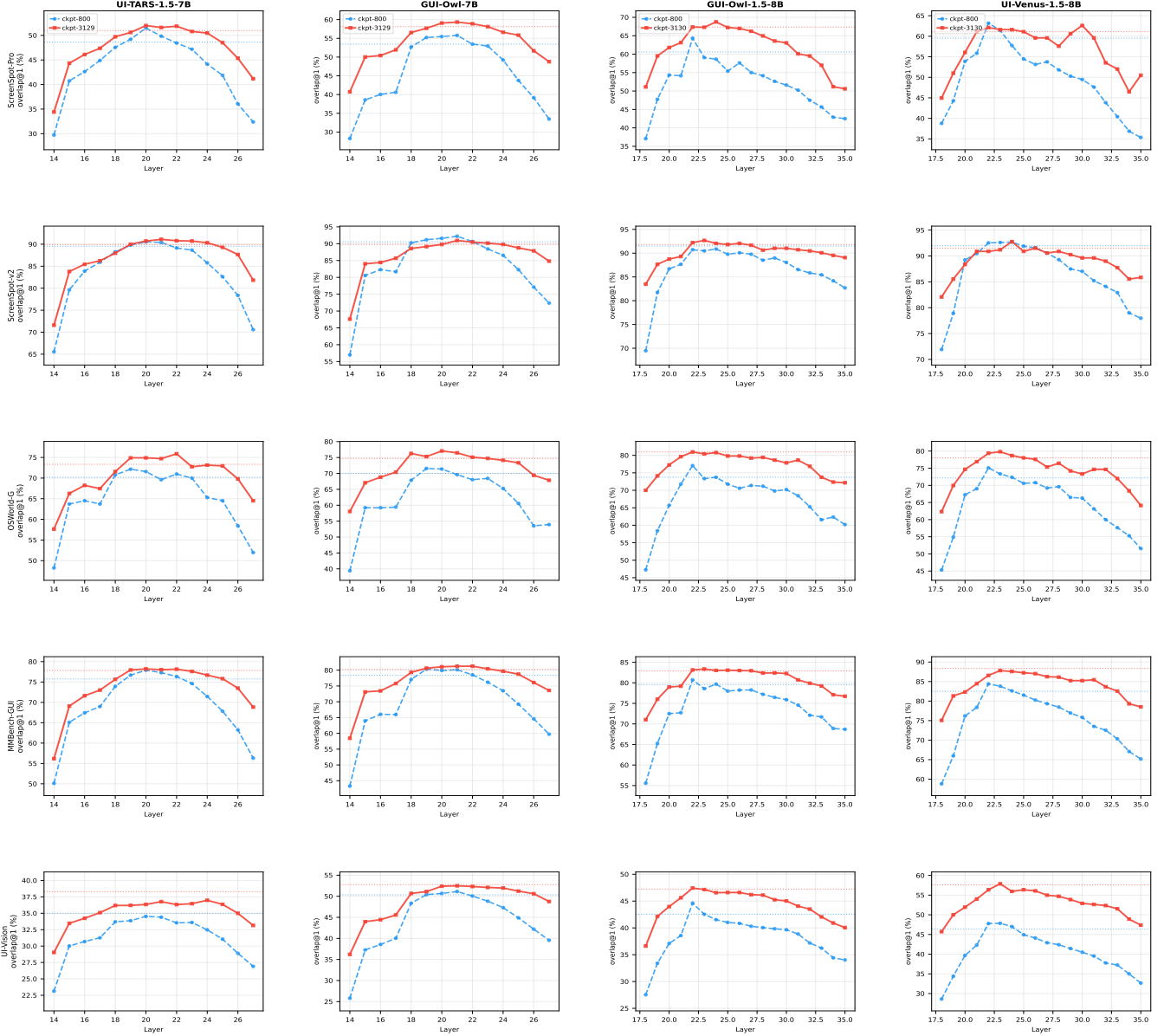


Figure 6: Per-layer overlap@1 of A7 CrossAttn probes across all four backbone models and five benchmarks. Each cell shows overlap@1 for a single probe layer evaluated independently (no cross-layer fusion). **Rows:** ScreenSpot-Pro (SS-Pro), ScreenSpot-v2 (SS-v2), OSWorld-G, MMBench-GUI, UI-Vision. **Columns:** UI-TARS-1.5-7B, GUI-Owl-7B, GUI-Owl-1.5-8B, UI-Venus-1.5-8B. **Blue dashed:** ckpt-800 (early training); **red solid:** last checkpoint (ckpt-3129 / ckpt-3130). **Horizontal dotted lines:** corresponding fusion head accuracy (the final ZWERGE-UI output). All four models exhibit an inverted-U spatial profile: accuracy peaks at intermediate layers and declines at the deepest layers, consistent with the coordinate serialization bottleneck (§3). The fusion head (dotted) tracks the best single-layer probe within 1–3 pp across nearly all panels (and exceeds it on a few), providing a deployable, sample-adaptive readout that does not require per-benchmark layer selection.

Per-Layer overlap@1 on ScreenSpot-Pro
(A7 CrossAttn Probe, each probe layer evaluated independently)

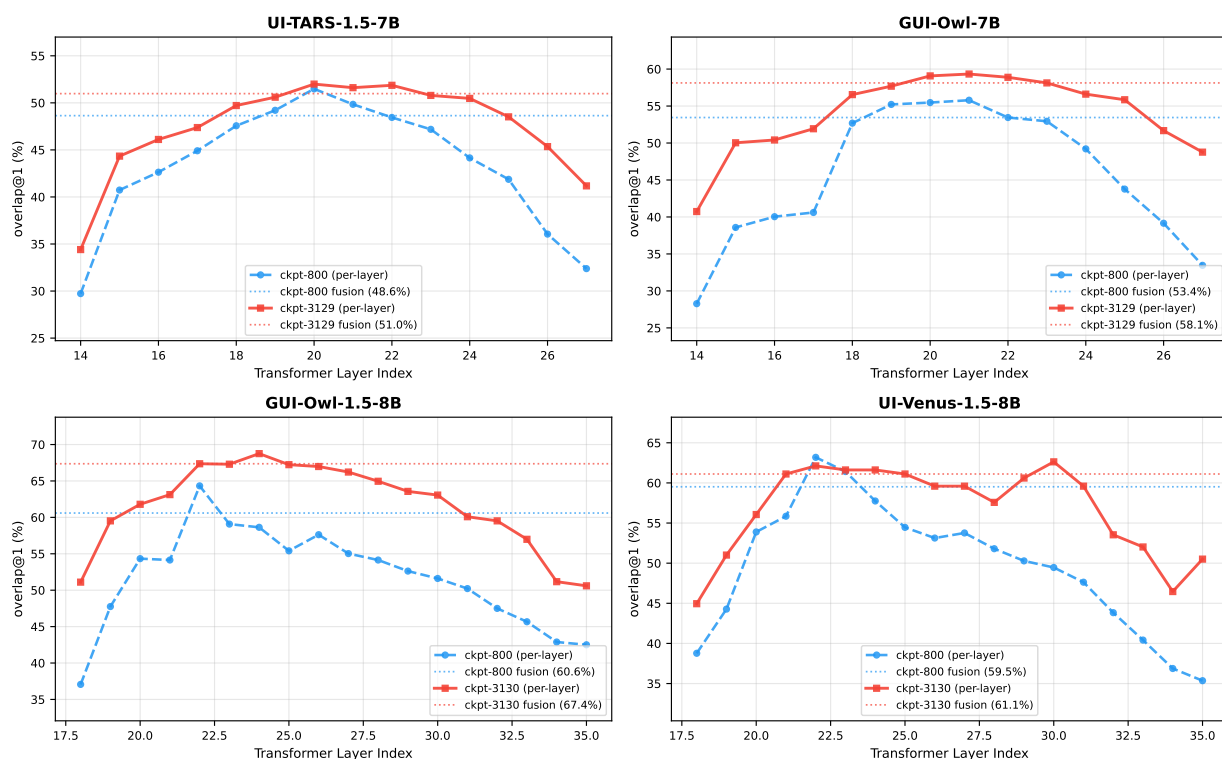


Figure 7: **Per-layer overlap@1 on ScreenSpot-Pro.** ScreenSpot-Pro is the most demanding benchmark (fine-grained targets on high-resolution desktop screenshots). The inverted-U profile is the sharpest here, with the deepest layers showing the steepest decline, reflecting that high-resolution targets require the most precise spatial encoding and are most sensitive to the serialization shift in deep layers. Note that the vertical scales differ across panels to emphasize the shape of the curve within each model.

Per-Layer overlap@1 on ScreenSpot-v2
(A7 CrossAttn Probe, each probe layer evaluated independently)

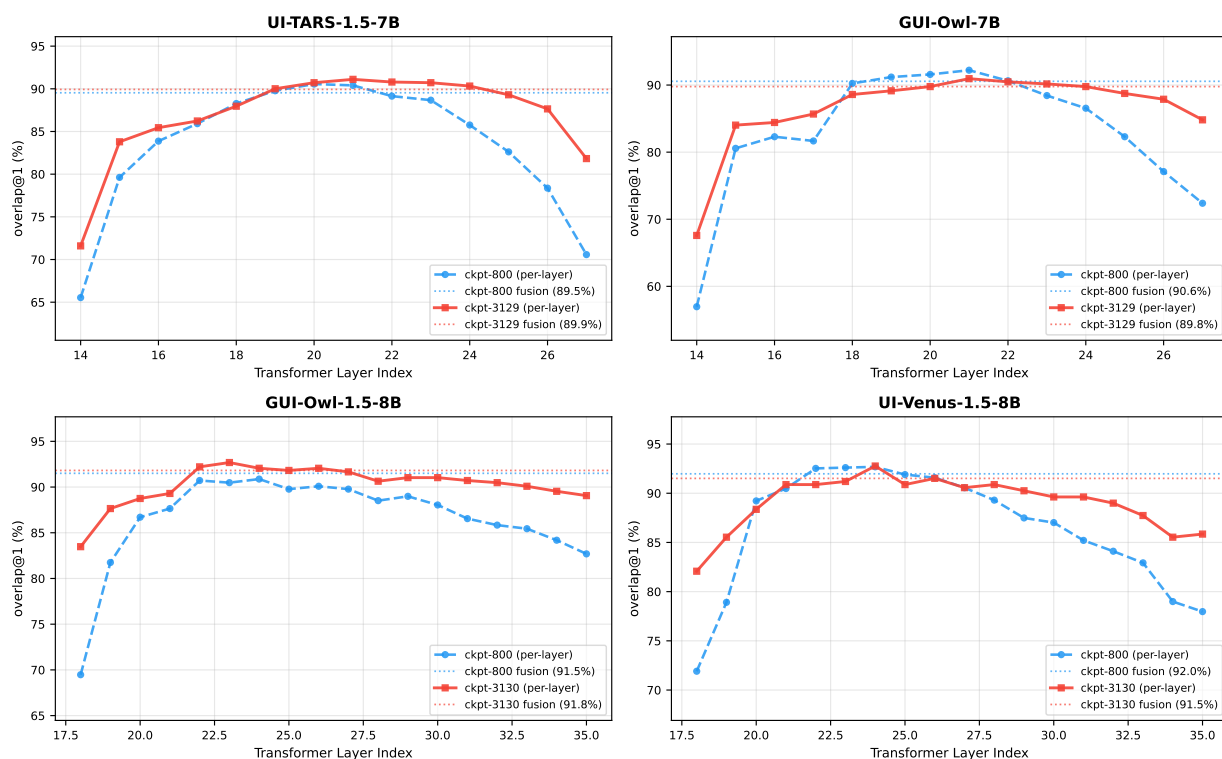


Figure 8: **Per-layer overlap@1 on ScreenSpot-v2.** ScreenSpot-v2 is a moderate-resolution benchmark (mobile, desktop, web) where overall accuracy is already high ($>87\%$ at ckpt-800). The inverted-U is shallower and the peak is broader compared to ScreenSpot-Pro, indicating that moderate-resolution grounding is less sensitive to which specific layer is read.

Per-Layer overlap@1 on OSWorld-G
(A7 CrossAttn Probe, each probe layer evaluated independently)

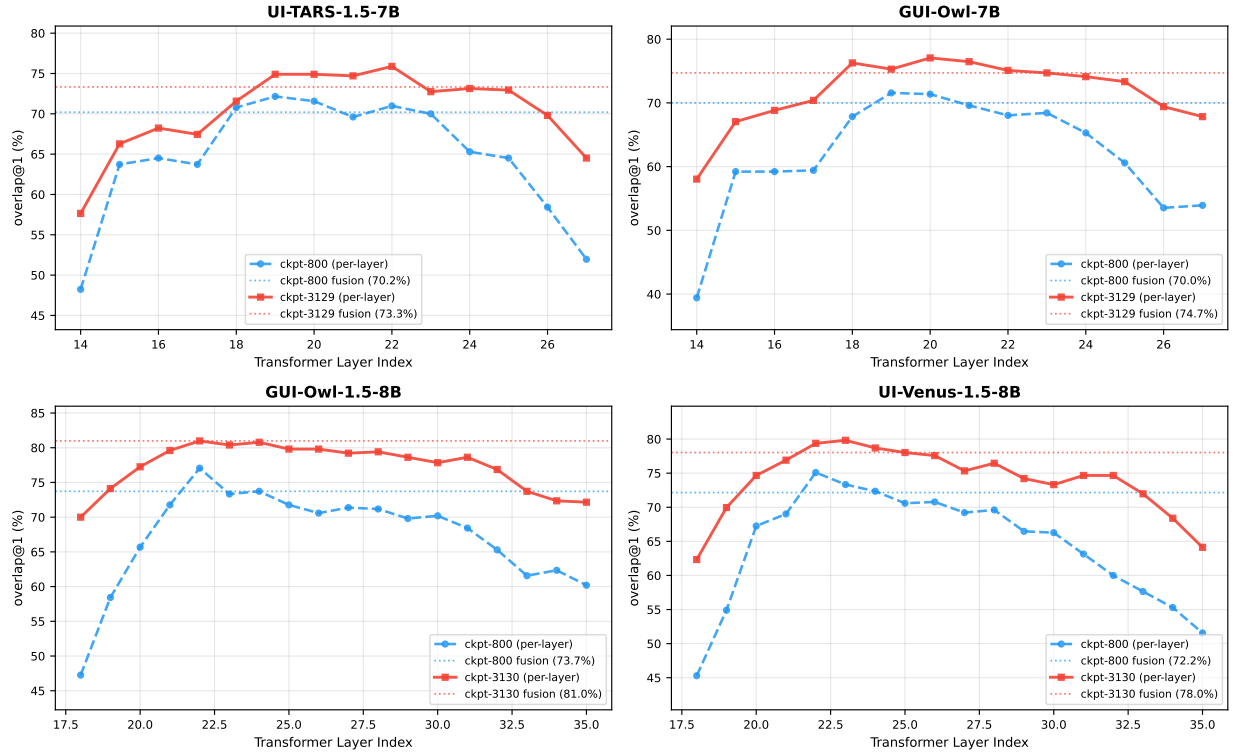


Figure 9: **Per-layer overlap@1 on OSWorld-G.** OSWorld-G provides real computer-task GUI samples with diverse task types (Text Matching, Element Recognition, Layout Understanding, Fine-grained Manipulation). The per-layer profile shows a similar inverted-U shape but with a plateau over a wider middle range ($\sim$ L18–L24 for 28-layer models), suggesting that layout-level spatial reasoning is distributed more broadly across intermediate layers than fine-grained element targeting.

Per-Layer overlap@1 on MMBench-GUI
(A7 CrossAttn Probe, each probe layer evaluated independently)

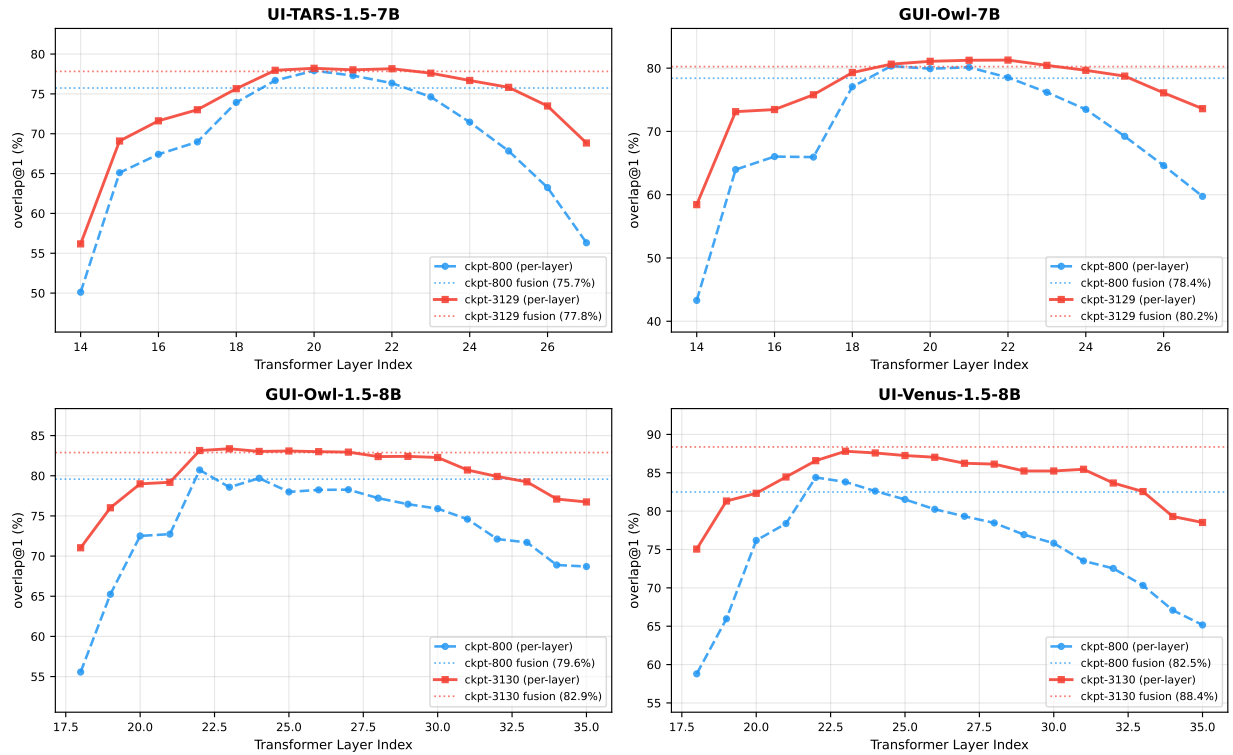


Figure 10: **Per-layer overlap@1 on MMBench-GUI.** MMBench-GUI covers six platforms (Windows, macOS, Linux, iOS, Android, Web). The per-layer curves are notably high even at ckpt-800 (blue dashed) for Qwen3-VL models (GUI-Owl-1.5-8B and UI-Venus-1.5-8B), reflecting that these models have strong cross-platform generalization already encoded in their intermediate representations. The gap between ckpt-800 and last checkpoint is smaller on MMBench-GUI than on ScreenSpot-Pro, suggesting that the per-layer signal saturates earlier on this benchmark.

Per-Layer overlap@1 on UI-Vision
(A7 CrossAttn Probe, each probe layer evaluated independently)

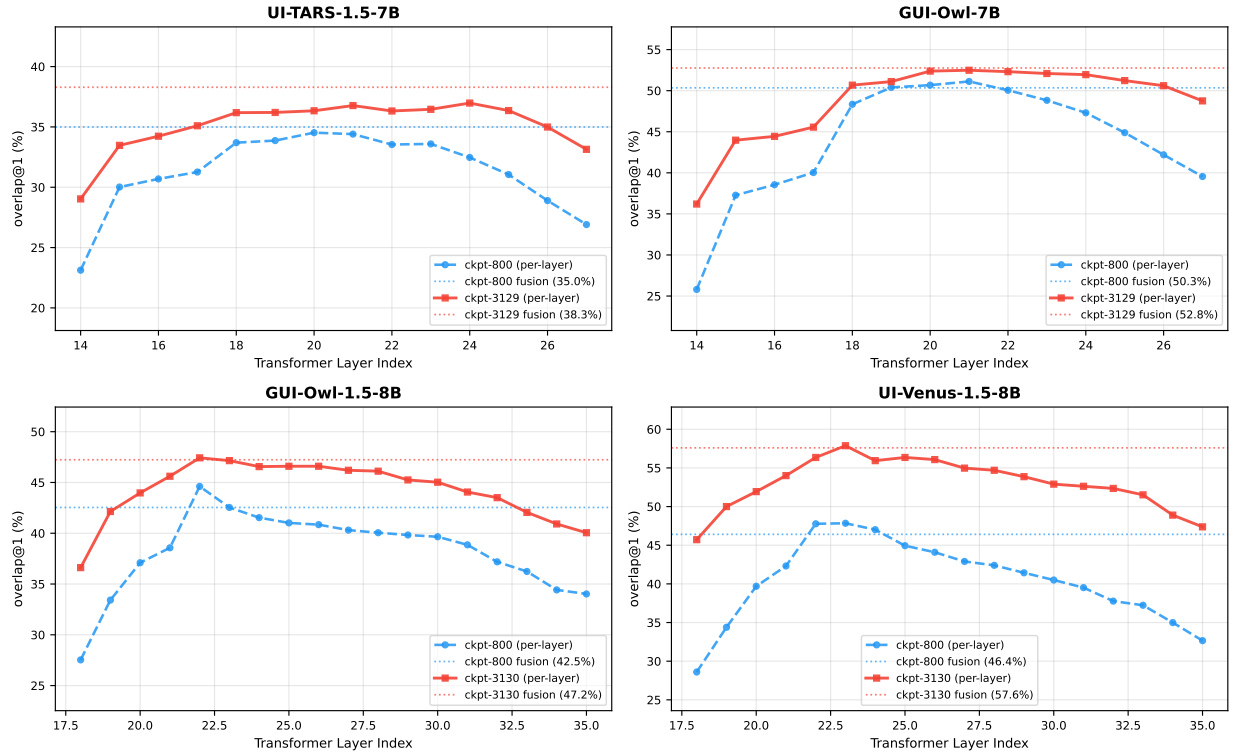


Figure 11: **Per-layer overlap@1 on UI-Vision.** UI-Vision decomposes grounding into Basic, Functional, Spatial, and Layout subtypes; accuracy here represents the weighted average over all 5,790 samples. Per-layer accuracy is substantially lower on UI-Vision than on other benchmarks, particularly for Qwen2.5-VL models at early training (ckpt-800 for UI-TARS-1.5-7B peaks below 30%), indicating that UI-Vision’s diverse task composition is harder for intermediate probes to solve with limited training. The inverted-U is still present but shallower and shifted earlier (peak around L18–L20), possibly because UI-Vision includes layout-level tasks that require higher-level semantic understanding.

Prompt for the UI-TARS-1.5 Model

```

1 You are a helpful assistant.
2
3 You are a GUI agent. You are given a task and your action history, with screenshots.
4 You need to perform the next action to complete the task.
5
6 ## Output Format
7 Action: ...
8
9 ## Action Space
10 click(start_box='<|box_start|>(x1,y1)<|box_end|>')
11
12 ## User Instruction
13 Task Instruction: {instruction}

```

Prompt for the GUI-Owl Model

```
1 You are a helpful assistant.
2 # Tools
3 You may call one or more functions to assist with the user query.
4 You are provided with function signatures within <tools></tools> XML tags:
5 <tools>
6 {
7   "type": "function",
8   "function": {
9     "name": "computer_use",
10    "description": "Use a mouse to interact with a computer screen."
11      "* The screen's resolution is ${height}x${width}."
12      "* Make sure to click any buttons, links, icons, etc with the cursor tip in the center of the
13        element. Don't click boxes on their edges unless asked.",
14    "parameters": {
15      "properties": {
16        "action": {
17          "description": "The action to perform. The available actions are:"
18            "* 'left_click': Click the left mouse button at a specified (x, y) pixel coordinate
19              on the screen.",
20          "enum": ["left_click"],
21          "type": "string"
22        },
23        "coordinate": {
24          "description": "(x, y): The x (pixels from the left edge) and y (pixels from the top edge)
25            coordinates to click.",
26          "type": "array"
27        }
28      },
29      "required": ["action"],
30      "type": "object"
31    }
32  }
33 </tools>
34 For each function call, return a json object with function name and arguments within <tool_call></tool_call>
35   XML tags:
36 <tool_call>
37 {"name": <function-name>, "arguments": <args-json-object>}
38 </tool_call>
39 ## User Instruction
40 Please complete the following tasks by clicking using 'left_click' function: {instruction}
```

Prompt for the GUI-Owl-1.5 Model

```
1  # Tools
2
3  You may call one or more functions to assist with the user query.
4
5  You are provided with function signatures within <tools></tools> XML tags:
6  <tools>
7  {"type": "function", "function": {"name": "computer_use", "description": "Use a mouse to interact with a
   computer.\n* The screen's resolution is 1000x1000.\n* Make sure to click any buttons, links, icons, etc
   with the cursor tip in the center of the element. Don't click boxes on their edges unless asked.", "
   parameters": {"properties": {"action": {"description": "The action to perform. The available actions are
   :\n* 'left_click': Click the left mouse button with coordinate (x, y) pixel coordinate on the screen.",
   "enum": ["left_click"], "type": "string"}, "coordinate": {"description": "(x, y): The x (pixels from the
   left edge) and y (pixels from the top edge) coordinates to click.", "type": "array"}}, "required": ["
   action"], "type": "object"}}}
8  </tools>
9
10 For each function call, return a json object with function name and arguments within <tool_call></tool_call>
    XML tags:
11 <tool_call>
12 {"name": <function-name>, "arguments": <args-json-object>}
13 </tool_call>
14
15 ## User Instruction
16 {instruction}
```

Prompt for the UI-Venus-1.5 Model

```
1  Output the center point of the position corresponding to the following instruction: {instruction}.
2
3  The output should just be the coordinates of a point, in the format [x,y].
```

Prompt for the UI-TARS-1.5 Retrofit Model

```
1  [System]
2  You are a GUI agent. You are given a task and a screenshot.
3  You need to perform the next action to complete the task.
4
5  ## Output Format
6
7  Action: ...
8
9  ## Action Space
10
11 click(start_box='<|ground|><|pointer_start|><|pointer_pad|><|pointer_end|>')
12
13 [User]
14 <image>
15 {instruction}
16
17 [Assistant]
18 click(start_box='<|ground|><|pointer_start|><|pointer_pad|><|pointer_end|>')
```

Prompt for the GUI-Owl-7B / GUI-Owl-1.5 Retrofit Model

```
1 [System]
2 # Tools
3
4 You may call one or more functions to assist with the user query.
5
6 You are provided with function signatures within <tools></tools> XML tags:
7 <tools>
8 {"type": "function", "function": {
9   "name": "computer_use",
10  "description": "Use a mouse and keyboard to interact with a computer, and take screenshots. This is an
    interface to a desktop GUI. You do not have access to a terminal or applications menu. You must click on
    desktop icons to start applications. * The screen's resolution is 1000x1000.\n* Make sure to click
    buttons, links, icons, etc with the cursor tip in the center of the element. Don't click boxes on their
    edges unless asked.",
11  "parameters": {"properties": {"action": {"description": "The action to perform. The available actions are:\n*
    'left_click': Click the left mouse button at coordinate (x, y) pixel coordinate on the screen.", "enum
    ": ["left_click"], "type": "string"}, "coordinate": {"description": "(x, y): The x (pixels from the left
    edge) and y (pixels from the top edge) coordinates to click. Required only by 'action=left_click'.", "
    type": "array"}}, "required": ["action", "coordinate"], "type": "object"}}}
12 </tools>
13
14 For each function call, return a json object with function name and arguments within <tool_call></tool_call>
    XML tags:
15 <tool_call>
16 {"name": "computer_use", "arguments": {"action": "left_click", "coordinate": [<|ground|><|pointer_start|><|
    pointer_pad|><|pointer_end|>]}}
17 </tool_call>
18
19 [User]
20 <image>
21 {instruction}
22
23 [Assistant]
24 <tool_call>
25 {"name": "computer_use", "arguments": {"action": "left_click", "coordinate": [<|ground|><|pointer_start|><|
    pointer_pad|><|pointer_end|>]}}
26 </tool_call>
```

Prompt for the UI-Venus-1.5 Retrofit Model

```
1 [User]
2 <image>
3 Output the center point of the position corresponding to the following instruction:
4 {instruction}.
5
6 The output should just be the coordinates of a point, in the format
7 [<|ground|><|pointer_start|><|pointer_pad|><|pointer_end|>].
8
9 [Assistant]
10 [<|ground|><|pointer_start|><|pointer_pad|><|pointer_end|>]
```



Figure 12: Successful cases of UI-TARS-1.5-7B on ScreenSpot-Pro

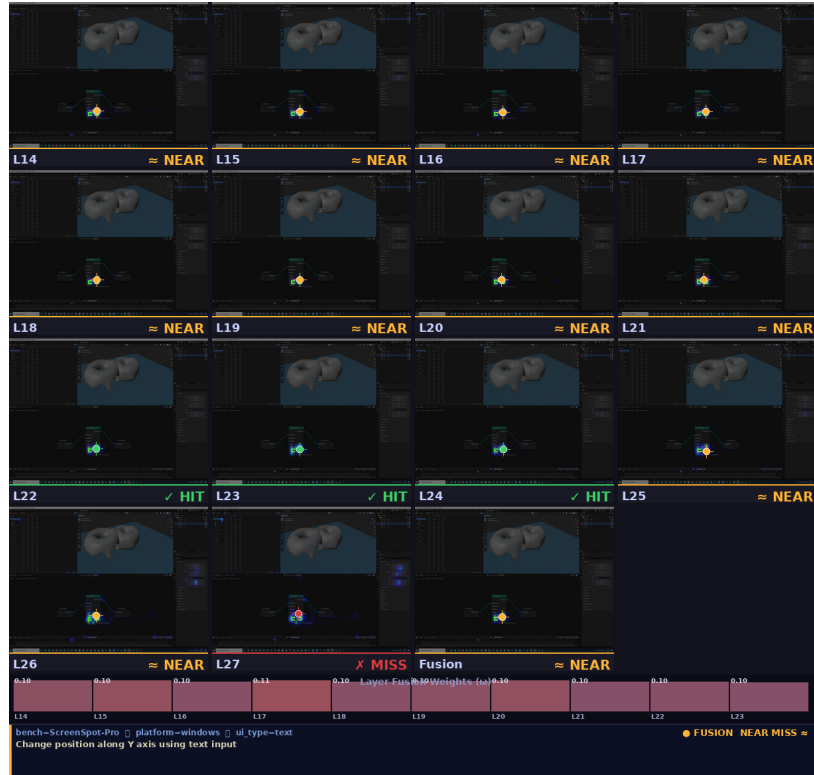


Figure 13: Failure cases of UI-TARS-1.5-7B on ScreenSpot-Pro



Figure 14: Successful cases of UI-Venus-1.5-8B on MMBench-GUI-L2



Figure 15: Failure cases of UI-Venus-1.5-8B on MMBench-GUI-L2

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